BRICK WORK · MILESTONE 4.0 · STATUS REPORT

# Prod runs on schema 4.0

Goal met on 28 Oct 2026 at 13:00. Every item has a Brick Work id, the catalogue comes from Rebrickable’s dump with no key, the collection was migrated from 3.3 with a proof, and Toni and lego-add use API v2.

**6 days**
23–28 Oct, planned 18–22 weeks

**93**
commits on main

**3,661**
tests, quarantine 0

**11 min**
prod down, 12:49–13:00

Gate #116 · ADRs 0016, 0017, 0018 · the plan as set follows after the report

DONE

# All twelve pieces closed

| Item | Piece | Closed |
|---|---|---|
| M.1 | Schema and connection | 23 Oct |
| M.2 | Catalogue builder | 24 Oct |
| M.3 | Shipped mappings | 23 Oct |
| M.4 | Figure mapping test | 23 Oct |
| M.5 | Acquired part lists | 25 Oct |
| M.6 | Copies and switching | 25 Oct |
| **M.7** | **Migration with proof** | 26 Oct |
| M.8 | Core modules on item_id | 27 Oct |
| M.9 | Test-suite rewrite | 27 Oct |
| M.10 | Screens, 36 frames approved | 28 Oct |
| **M.11** | **API v2, MCP, both clients** | 28 Oct |
| **M.12** | **Onboarding, checks, cutover** | 28 Oct |

Achieved highlights: a migration proof that passed identically on the Mac, on test and on prod; 188 API v2 operations with an MCP that takes numbers; a fresh install that reaches a checkable set with no key.

AGAINST THE PLAN

# What changed, and why

| Change | Why |
|---|---|
| 6 days, not 18–22 weeks | One piece at a time: gate, Codex review, real-data check |
| Toni App Lego tab added | About 45 v1 calls, missing from #113, found in M.11f |
| Catalogue entry gains theme | Migrated items lack properties until jobs re-fetch them |
| Runbook in rasp-infra | It is host glue, not app documentation |
| No BrickLink lists in CAT | The configuration the rehearsal proved; they load later |
| Pi first-import timing | Not measured; filed as #126 |

DECISIONS

# Three decision records

ADR 0016

## Data model 4.0

Brick Work ids for every item, a catalogue from Rebrickable’s dump, providers linked with a confidence, one migration with a proof.

ADR 0017

## API v2 and its MCP

v1’s resources with the 4.0 identity. Number lookups default to BrickLink; ambiguous is 409; merged is 307. MCP tools take numbers, never ids.

ADR 0018

## The cutover

One pinned catalogue, a rehearsal from a prod backup, the clients in the window, rollback by rename and image tag.

THE GATE · #116

# Each check, with its evidence

| Check | Evidence |
|---|---|
| M.1–M.12 closed | #103–#114 closed, the last three on 28 Oct |
| Proof PASS, rehearsed on test | #114: lego-test from the prod backup, pinned catalogue |
| Prod on 4.0, 3.3 file kept | #115: prod-1c23570; data/brickwork-3.3.db, 196 copies |
| Toni and lego-add on v2 | House API on v2, lego-add lookup and an Undo write on prod |
| Catalogue in Borg, checked | brick-work-2026-10-28_13:03: catalogue.db, dump ok |
| Close-out | This report; PDF in docs/milestones/4.0-data-model.pdf |

RETROSPECTIVE

# What worked, what didn’t, what changes

### Went well

Real-data checks found what tests missed: a price stored under the wrong colour, and 16% attached without the mapping layer.

The rehearsal found five runbook defects; prod’s numbers then matched it exactly.

Rollback rehearsed: 9 s to healthy.

### Went badly

The client list was incomplete: the Lego tab came late, and an env template kept /api/v1 (410 for a minute).

GitHub comments went out as Ulf, not the bot.

The cutover-day Borg run pruned the 02:01 archive.

### Improve

**Applied:** the runbook checks deployed client config and names the Borg prune.

**Sent to the wiki-agent:** a client inventory before a breaking API, a bot-only GitHub hook, a venv for worktree gates.

CARRIES OVER

# Into milestone A and the backlog

| Item | Where |
|---|---|
| Re-plan A against 4.0 | Before A starts (A's deck, #83, #88) |
| Properties, pictures, prices | Re-fetched by jobs; Brickset about 4 days |
| 30 manual identity decisions | Redo from migrate4-report/manual-decisions.csv |
| Site API page still v1 | #125 |
| Pi first-import timing | #126 |
| Adding a set in the UI | #121, works on 4.0 on test; Ulf to confirm on prod |
| Retire the old BrickTracker | Ulf, no earlier than 17 Nov |

BRICK WORK · MILESTONE 4.0

# Data model 4.0

Goal: Brick Work runs on schema 4.0 in prod. Every set, part, minifigure and colour has a Brick Work id; the catalogue comes from Rebrickable’s dump with no key; the collection is migrated from 3.3 with a proof; API v2 replaces v1.

Rough length: 18–22 weeks · Before milestone A · Gate: issue #116 · ADR 0016

WHY

# BrickLink numbers cannot carry a catalogue that comes from Rebrickable

### Today, 0.5

BrickLink’s type and number is the identity of everything. What cannot be translated becomes a “?” item. Sticker sheets, legs whose BrickLink id depends on colour, and assemblies have no place.

### After 4.0

Brick Work gives every item its own id. BrickLink, Rebrickable and Brickset link to it with a confidence: matched, inferred, manual or to check, scoped to a colour where needed.

Spike #102 settled the source: Rebrickable’s dump is free to use with acknowledgement, BrickStore’s archive of BrickLink data has no shown permission. The identity touches 54 key references in 40 tables, so the whole schema is re-derived.

WHAT CHANGES FOR THE USER

# A reference book and a notebook

| Area | In 4.0 |
|---|---|
| Two files | catalogue.db is rebuilt from the dump; brickwork.db holds the collection and is never touched by a rebuild |
| Adding a set | Offline and instant from the catalogue; a part-list picker when there is a choice |
| Part lists | Rebrickable by default; BrickLink’s list per set or by a global switch; “version n of m” |
| Keys | None needed to start; BrickLink adds prices, order import and its part lists |
| Migration | Collection, prices, keys and tokens, preferences; history stays in the old file |
| API | v2 by Brick Work id with number lookups; v1 answers 410; Toni and lego-add switch |

THE WORK

# Twelve pieces, about 20 weeks

| Item | Piece | Weeks |
|---|---|---|
| M.1 | Schema and connection | 1 |
| M.2 | Catalogue builder | 2.5 |
| M.3 | Shipped mappings | 1 |
| M.4 | Figure mapping test | 0.75 |
| M.5 | Acquired part lists | 0.75 |
| M.6 | Copies and switching | 1 |

| Item | Piece | Weeks |
|---|---|---|
| M.7 | Migration with proof | 2 |
| M.8 | Core modules on item_id | 4.5 |
| M.9 | Test-suite rewrite | 2.5 |
| M.10 | Screens, frames first | 2 |
| M.11 | API v2 and clients | 1.5 |
| M.12 | Onboarding, checks, cutover | 0.75 |

Plus 1 week of contingency. M.1, M.3 and M.4 can start together. The least certain lines are the core modules, the test rewrite and BrickStore’s file format.

THE GATE · #116

# Prod runs on 4.0, proven against 3.3

- M.1–M.12 closed
- Migration proof PASS on a copy of prod on test, rehearsed with the cutover catalogue
- Cutover decided (#115), prod on 4.0, the 3.3 file kept beside it
- Toni and lego-add working against API v2
- Catalogue file in Borg, first archive checked
- Close-out: this deck becomes the milestone’s status report